

Answer Key & Teacher Notes

Session 1 – Understanding Ratios • Session 2 – Ratios and Tape Diagrams

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Understanding Ratios

TALK PROMPTS

- Does order matter in a ratio? Is $4 : 6$ the same as $6 : 4$?
- Why does $4 : 3$ describe the same garden as $24 : 18$ (Problem 4)?

PARTNER TASK

Band Sections. In Problem 6, one partner writes trumpets to flutes in simplest form ($4 : 7$); the other writes drums to all band members ($3 : 14$). Explain why you add all the groups for a part-to-whole ratio.

WATCH FOR

- Writing the terms in the wrong order (dogs to cats written as cats to dogs).
- Using one group as the whole in a part-to-whole ratio instead of adding both groups.
- Dividing only one term when simplifying ($12 : 18$ written as $6 : 18$).

SESSION 2 • TEACHER NOTES

Ratios and Tape Diagrams

TALK PROMPTS

- How does dividing the total (25) by the total parts (5) help you find each ingredient?
- Multiply each row by the value of one box. Do your two amounts add back to the total?

PARTNER TASK

Garden Beds. In Problem 8 (beans to corn $5 : 3$, 96 seeds), one partner finds the total parts and one box (12); the other finds each amount (60 beans, 36 corn) and checks that the sum is 96.

WATCH FOR

- Dividing the total by one ratio term instead of the total parts.
- Using the unit value as the answer without multiplying by the group's number of boxes.
- Forgetting to check that the shares add back to the total.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.1	1	2	1	2	2	2	2	2	3	1	1
S1 • Form B	6.RP.A.1	1	2	1	2	2	2	2	2	3	1	1
S2 • Form A	6.RP.A.1	2	1	1	2	2	2	2	2	3	1	1
S2 • Form B	6.RP.A.1	2	1	1	2	2	2	2	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

ratio: A comparison of two **amounts**.

term: One of the two **numbers** in a ratio.

part-to-part: Compares one group with **another** group.

part-to-whole: Compares one group with the **total**.

- Write a ratio in the **order** the question asks.
- For part-to-whole, **add** the groups to find the whole.
- To simplify, divide both terms by the same **number**.

Example: All books: $6 + 9 = 15$. Math first: $6 : 15$. Divide both terms by **3**: $2 : 5$. Answer: Math to all books = **2 : 5**

YOU TRY

1. a. $10 : 15 = 2 : 3$ b. $15 : 25 = 3 : 5$
2. $4 : 7$

EXIT TICKET

1. a. $6 : 9 = 2 : 3$ b. $9 : 15 = 3 : 5$
2. $8 : 20 = 2 : 5$
3. **A** ($2 : 5$)

PRACTICE FORM A

1. a. $4 : 6 = 2 : 3$ b. $2 : 12 = 1 : 6$ c. $6 : 4 = 3 : 2$ d. $4 : 12 = 1 : 3$
2. a. 12 to 8, $12 : 8, 12/8 = 3 : 2$ b. $8 : 20 = 2 : 5$
3. $15 : 10 = 3 : 2$
4. $24 : 18 = 4 : 3$. Yes, the helper is correct.
5. mystery = $30 - 12 = 18$; $18/30 = 3/5$
6. a. $8 : 14 = 4 : 7$ b. $6 : 28 = 3 : 14$
7. a. one box = 5, so blue = $3 \times 5 = 15$ marbles b. all = $5 + 15 = 20$; $5 : 20 = 1 : 4$
8. a. $12 : 16 = 3 : 4$ b. $16 : 28 = 4 : 7$ c. No. Two more votes for Ana make $14 : 16 = 7 : 8$, not $1 : 1$.
9. Devon reversed the order. Dogs come first, so dogs to cats is 8 to 5.
10. **B** ($2 : 3$)
11. **B** ($5 : 12$)

Self-check word RATIOS

PRACTICE FORM B

1. a. $6 : 10 = 3 : 5$ b. $4 : 20 = 1 : 5$ c. $10 : 6 = 5 : 3$ d. $6 : 20 = 3 : 10$
2. a. 15 to 6, $15 : 6, 15/6 = 5 : 2$ b. $6 : 21 = 2 : 7$
3. $14 : 6 = 7 : 3$
4. $21 : 28 = 3 : 4$. No, the helper reversed the order.
5. mystery = $35 - 15 = 20$; $20/35 = 4/7$
6. a. $6 : 10 = 3 : 5$ b. $4 : 20 = 1 : 5$
7. a. one box = 6, so blue = $4 \times 6 = 24$ marbles b. all = $6 + 24 = 30$; $6 : 30 = 1 : 5$
8. a. $12 : 18 = 2 : 3$ b. $18 : 30 = 3 : 5$ c. No. Four more votes for Ana make $16 : 18 = 8 : 9$, not $1 : 1$.
9. Maya reversed the order. Guinea pigs come first, so guinea pigs to rabbits is 7 to 4.
10. **D** ($2 : 3$)
11. **C** ($4 : 13$)

Self-check word SIMPLE

GUIDED NOTES

tape diagram: A drawing with **equal** boxes that shows a ratio.

unit (one box): The amount inside **one** box.

total parts: All the boxes **added** together.

- Add the ratio terms to find the **total** parts.
- One box = total $\div$ **total parts**.
- Multiply one box by each **term** to find each share.

Example: Total parts: $3 + 4 = 7$. One box: $28 \div 7 = 4$ stickers. Ana: $3 \times 4 = 12$. Ben: $4 \times 4 = 16$. Answer: Ana gets **12** and Ben gets **16**. Check: $12 + 16 = 28$

YOU TRY

1. a. 8 b. 9 c. 10
2. 7 parts; $\$56 \div 7 = \8 ; \$24 and \$32

EXIT TICKET

1. a. 7 b. 7 c. 8
2. 9 parts; $36 \div 9 = 4$; 8 and 28
3. **A** (total parts)

PRACTICE FORM A

1. a. $2 + 3 = 5$ parts b. $25 \div 5 = 5$ cups c. granola $2 \times 5 = 10$ cups; cranberries $3 \times 5 = 15$ cups
2. $20 \div 5 = 4$; bananas $2 \times 4 = 8$
3. $35 \div 5 = 7$; blue $4 \times 7 = 28$ cups
4. $5 + 3 = 8$ parts; $32 \div 8 = 4$; goldfish $5 \times 4 = 20$
5. $42 \div 6 = 7$ counselors
6. $4 + 5 = 9$; $\$45 \div 9 = \5 ; \$20 and \$25
7. a. $15 \div 5 = 3$ oz b. 7 parts $\times 3 = 21$ oz
8. a. $96 \div 8 = 12$; beans 60, corn 36 b. $60 - 36 = 24$ c. 1 : 1 (48 and 48)
9. She divided by one term, not the total parts. Total parts = 7, one box = $28 \div 7 = 4$, so sugar = $2 \times 4 = 8$ cups.
10. **B** (6)
11. **C** (20)

Self-check word **DIAGRAM**

PRACTICE FORM B

1. a. $3 + 2 = 5$ parts b. $30 \div 5 = 6$ cups c. oats $3 \times 6 = 18$ cups; raisins $2 \times 6 = 12$ cups
2. $21 \div 7 = 3$; bananas $3 \times 3 = 9$
3. $40 \div 5 = 8$; white $1 \times 8 = 8$ cups
4. $3 + 5 = 8$ parts; $40 \div 8 = 5$; guppies $5 \times 5 = 25$
5. $60 \div 12 = 5$ counselors
6. $3 + 4 = 7$; $\$56 \div 7 = \8 ; \$24 and \$32
7. a. $20 \div 4 = 5$ oz b. 7 parts $\times 5 = 35$ oz
8. a. $96 \div 12 = 8$; carrots 56, peas 40 b. $56 - 40 = 16$ c. 1 : 1 (48 and 48)
9. He divided by one term, not the total parts. Total parts = 7, one box = $42 \div 7 = 6$, so rice = $3 \times 6 = 18$ cups.
10. **B** (4)
11. **A** (5)

Self-check word **DIVIDED**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • UNDERSTANDING RATIOS

1. a. $2 : 3$ b. $3 : 2$ c. $6 : 15 = 2 : 5$ d. $9 : 15 = 3 : 5$
2. a. $2 : 3$ b. $5 : 2$ c. $3 : 7$
3. Part-to-whole. The 15 counts all the players on the team (8 girls + 7 boys).
4. Aquarium votes: $24 - 14 = 10$. Museum to aquarium = $14 : 10 = 7 : 5$. For every 7 museum votes, there are 5 aquarium votes.
5. Marcus reversed the order: $2 : 5$ is cats to dogs ($4 : 10$). Dogs to cats is $10 : 4 = 5 : 2$.

SHEET 2 • RATIOS AND TAPE DIAGRAMS

1. a. $3 + 5 = 8$ boxes; $48 \div 8 = 6$ beads b. Red: $3 \times 6 = 18$; Blue: $5 \times 6 = 30$ ($18 + 30 = 48$)
2. a. 18 and 24 b. 12 and 54
3. a. $12 \div 3 = 4$ b. $4 \times 4 = 16$ melon pieces c. $12 + 16 = 28$ pieces
4. a. Theo has $7 - 2 = 5$ more boxes, so 5 boxes = 35 and one box = 7 stickers b. Maya $2 \times 7 = 14$; Theo $7 \times 7 = 49$; total $14 + 49 = 63$ stickers
5. He divided 45 by 5 (one term) instead of by the total parts, so his shares add to 81, not 45. Total parts = $4 + 5 = 9$; $45 \div 9 = 5$ per box; shares are $4 \times 5 = 20$ and $5 \times 5 = 25$.
6. **D** (\$25)
7. **B** (8)